MMCS Group Project · 2026 · Tier 1

Recovering the value of surplus renewable generation

Battery or electrolyser, at what scale, and under which contract

PREPARED BY	VERSION	DATA WINDOW	DECISION OWNER
Group 15	**1.0 — 5 November 2026**	**1 Jan 2020 – 17 Sep 2026**	**Portfolio board**

HEADLINE RESULTS

+£8.57m	20 MW	2.6 yr	0.07%
annual value of the 20 MW electrolyser module	electrolyser; no battery is economic	payback before stack replacement	of current electricity sales diverted to hydrogen

Decision

Build **20 MW of PEM electrolysis** and **no battery**, conditional on a hydrogen offtake of at least 2.2 t/day at £7.80/kg *and* on the operating mandate to divert 0.07% of current electricity sales. Capital is £29.2m, the annual increment £8.57m, the payback 2.6 years before stack replacement. Under a current-generation-first obligation the same asset loses £1.07m a year, so the mandate — not the price — is the binding condition.

1 Executive summary

The client's portfolio generated 215.7 TWh in 2023 against 245.8 TWh of demand: 87.5% of demand can be served, and 549.9 GWh of generation is stranded on the 29 days when supply exceeds demand. Of the two candidate assets, the brief's 1.5 MW/3 MWh battery is not economic at any non-negative capital cost, and a 20 MW electrolyser adds £8.57m a year — positive in every year replayed from 2020 to 2026, weakest in 2021 at £6.29m.

The mechanism is not the surplus but the contract. The module produces at most 8.59 t/day of hydrogen, just under the buyer's 10 t/day cap, so capacity is set by the offtake ladder rather than by the stranded energy. Of the 171.8 GWh the electrolyser consumes, 13.9 GWh (8.1%) is electricity that would otherwise be unused and 157.9 GWh (91.9%) is diverted from direct sale: the honest headline is that eight per cent is avoided waste and ninety-two per cent is an alternative use of saleable electricity.

What the reader has to accept: daily averages, perfect foresight, constant 2020 money. The offtake contract, not the capital cost, is what would move the answer.

The assumption that carries the answer

The recommendation holds while the client may divert about **0.07% of current sales** (157.9 GWh). A 99.9% supply floor still permits the full £8.57m; a 99.99% floor permits 35 GWh of hydrogen and £1.04m; a 100% floor excludes the investment entirely.

2 Problem, boundary and data

The decision is how much battery and electrolyser capacity to add, and how much unused electricity it can capture. Three numbers frame it: 215.7 TWh generated, 245.8 TWh demanded,

549.9 GWh stranded on 29 days. The first two say any new load is an alternative use of saleable electricity; the third — 0.26% of a year's generation — is the ceiling on the physical prize.

We use the five supplied tables in the independently repaired release (2,452 days, 1 January 2020 to 17 September 2026), keeping the course originals as a sensitivity: the original price is about half the repaired value and the original demand is truncated on the last day of each weekly query. Supplied generation plus the specified wind farm is a *constructed* system, not observed curtailment, and demand is the absorption cap — no purchases, no additional export route, no curtailment payment. The analysis is daily and perfect-foresight, without network, intraday, ancillary or hydrogen storage; money is constant 2020 GBP at 5% real; the battery is the brief's 1.5 MW/3 MWh module (90% round-trip, 20–80% window); the electrolyser is 20 MW PEM at 55.9 kWh/kg; hydrogen is valued at £7.80/kg in 2020 money, 1.4% below the brief's £9.5/kg benchmark once that is deflated to the same year (£7.91). Every £/kg figure below is 2020 money unless the year is named.

Data repair. The repaired release aligns five series to 2,452 whole UTC days (1 January 2020 to 17 September 2026, 2026 partial), rebuilds the partial demand windows stored on weekly query-end dates (340 of the 351 largest demand changes fall on a Tuesday), removes zero-volume price records and keeps the two non-positive price days as observed, rebases every flow to 2020 GBP with monthly CPI, and flags 523 days as non-strict (85 in 2023); the strict 2023 subset is re-run as a sensitivity. Source: the release's repair log [1].

3 Model

For day t, with generation G_t, demand D_t and market price p_t, the model chooses direct sale q_t, battery charge c_t and discharge d_t, electrolyser input h_t, unused energy u_t, operating modules z_t and installed units n_B, n_H:

$$\max\ \Pi = \frac{1}{Y}\sum_t \left[\frac{p_t\,(q_t + d_t)}{1000} + \frac{v\,h_t}{55.9}\right] - a_B\,n_B - a_H\,n_H \tag{1}$$

subject to

$$q_t + c_t + h_t + u_t = G_t, \qquad q_t + d_t \le D_t, \tag{2}$$

$$0.096\,z_t \le h_t \le 0.48\,z_t \le 0.48\,n_H, \qquad h_t \le 55.9\,L/10^6, \tag{3}$$

$$\textstyle\sum_t d_t = 0.9\sum_t c_t, \tag{4}$$

where p_t is £/MWh, v the hydrogen value in £/kg, 55.9 kWh/kg the electrolyser's specific consumption, L the offtake cap in tonnes per day, a_B, a_H the annualised charges from (5) and Y the days in the year. Policy A allocates each day's generation economically; policy B adds $q_t = \min(G_t, D_t)$, so current generation serves current demand first. B is the operating reality, not a preference.

Annualised charges use one convention for both assets,

$$a_j = F_j + \frac{i}{1-(1+i)^{-N_j}}\Big(K_j + \sum_{k\in R_j}\frac{C_{jk}}{(1+i)^k}\Big), \qquad j \in \{B, H\}, \tag{5}$$

with $i = 5\%$, $N_B = 15$, $N_H = 30$, $R_B = \emptyset$ and $R_H = \{11, 22\}$: the electrolyser recovers £0.52m of fixed cost, £1.90m of capital and £0.59m of stack renewals, £3.01m a year; the battery £22.3k of fixed cost and £74.1k of capital, £96.4k a year. The £3.01m is the third line of Table 2.

The annual battery balance (4) is the Tier 1 time simplification and is an upper bound on battery value; no selected configuration uses a battery, so it cannot change the recommendation. One design year has 2,556 rows and 2,192 columns, solves to proven optimality in under 0.1 s with Xpress 9.9.1, and 185 optimisation cases plus 2 rule-based comparators were run; twenty independent checks pass.

4 Results

Table 1. Configuration comparison, 2023, annual increment against the matching no-investment case, £m/yr.

Policy	No invest.	Battery	H_2 only	Sell-first
A: economic allocation	0	0	**+8.57**	−1.07
B: current generation first	0	0	0	−1.07

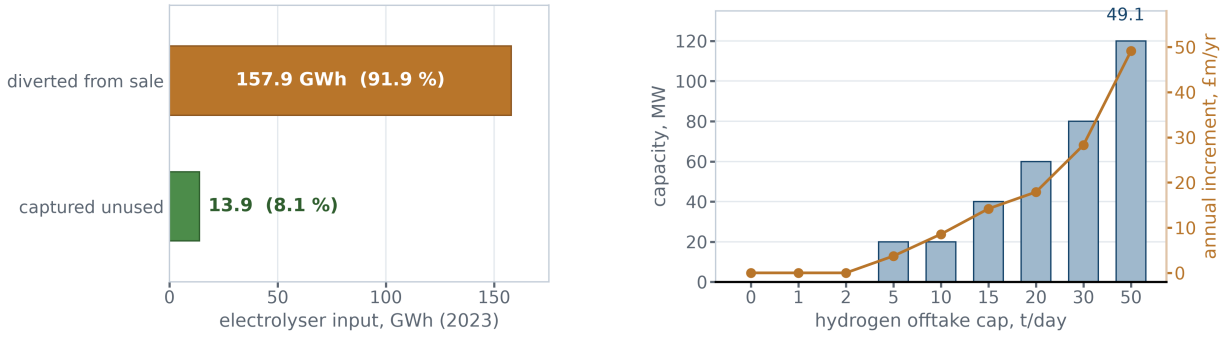
The battery is never selected, so joint choice equals the electrolyser case; zero under B means the asset is not run, and the last column is a fixed 20 MW module on a sell-first rule.

Table 2. Where the £8.57m comes from, 2023, £m/yr.

Line	£m/yr	Note
Hydrogen revenue	23.97	3,074 t over the year at £7.80/kg
Electricity revenue forgone	−12.39	157.9 GWh at the daily price
Fixed O&M; stack reserve; capital charge	−3.01	0.52 + 0.59 + 1.90
Net annual increment	**+8.57**	30-year NPV +£132m

The NPV is a scenario value, not a forecast: no price decay, no degradation, no terminal value.

The battery is never selected. At zero capital cost and zero fixed O&M a 1.5 MW unit is worth £13.25k a year, while the primary-scheme fixed O&M alone is £22.3k per unit, so no unit is economic at any non-negative capital cost; on the lower table-based O&M the break-even capital cost is £7.0k/MWh, about 37 times below the £256k/MWh equivalent of the brief's €288k/MWh. Because the annual-pool model overstates battery value, the conclusion is robust to the relaxation.



(a) attribution of electrolyser input

(b) capacity and value against the offtake cap

Figure 1. Left: the 171.8 GWh electrolyser input splits into 13.9 GWh of otherwise-unused generation and 157.9 GWh diverted from sale. Right: the capacity ladder is driven by the contract, not by the surplus.

Source: `results/results.json`, package `tier1/20261105-v1`. Both panels are redrawn from the package figures at printed size for legibility; the data are unchanged.

On the 29 surplus days the renewable share of generation is 52.7%, so a pro-rata attribution of the captured energy is about 7.3 GWh; at daily resolution the strict renewable bound is 0 to 13.9 GWh. The capacity ladder follows the contract: 0 MW at 2 t/day or less, 20 MW at 5–10 t/day, 40 MW at 15, 60 MW at 20, 80 MW at 30 and 120 MW at 50 t/day (+£49.1m a year on £175m of capital). At a 10 t/day cap the investment switches on at a netback of £4.62/kg; at £7.80/kg it needs 2,230 kg/day, and at £9.5/kg (2020 money, £11.4 at 2023 prices), 1,717 kg/day. Capacity stays at 20 MW anywhere between £7.80 and £9.50, so the 1.4% gap to the deflated brief benchmark does not move the decision. Fixing 20 MW and replaying each year keeps policy A positive throughout,

from +£6.29m (2021) to +£16.78m (2020), while policy B loses money in 2023 and 2024 (−1.07m and −1.00m). The spread is business risk, and it is the freedom to allocate generation that the asset depends on.

5 What changes it, and what to do

Electrolyser CAPEX of £0.82m–£1.65m/MW gives +£9.41m to +£8.33m with capacity unchanged; a 0–12% real discount rate gives +£9.44m to +£6.99m; the strict 2023 days give +£8.69m; dropping the wind farm +£8.13m; the unrepaired price +£15.19m, which is why it is not used. None of these moves the decision — the supply obligation does.

1. No battery in Tier 1 at the brief's cost. Re-open it only with intraday or ancillary revenue, or a capital cost near £7k/MWh.
2. Negotiate the hydrogen contract first: at least 2.2 t/day at £7.80/kg, or 1.7 t/day at £9.5/kg. Below a £4.62/kg netback, do not build.
3. Build 20 MW only with the contract *and* the operating mandate to divert about 0.07% of sales.
4. Stage the expansion: 20 MW for a 10 t/day contract, 120 MW for 50 t/day. Do not pre-build, and report the capture honestly: 13.9 GWh of avoided waste and 157.9 GWh diverted from sale.

6 Validation and limitations

Every stored plan closes its daily energy balance and its demand, charge, discharge, module and offtake caps; integer units and the annual battery balance hold, and revenues and costs are recomputed from the plans. The reference cases reproduce the independently built Baseline 1104 package (A0 17,139.332; 20 MW +8.572; fixed module −1.065; 2024 +11.951) — arithmetic agreement, not an endorsement of shared assumptions. A third, separately written Xpress implementation of the same boundary [7] matches all 2,452 input days and the 18 matched 2023–2025 cases (largest annual difference £49.54, 0.005%) and solves the 40 MW counterfactual at +£7.47m, below 20 MW.

Limitations, measured rather than conceded: daily averages remove intraday value, so the battery figure is an upper bound; the annual battery balance is a relaxation; dispatch is perfect-foresight; the hydrogen value is a reference, not a contract; CAPEX is a modern midpoint while O&M and stack renewals use older UK proxies; no degradation, hydrogen storage, network constraint or ancillary market; 523 of the 2,452 days carry a non-strict flag, so the renewable attribution is conditional on the repair; 2026 is partial.

Reproducibility and references

[1] Data: Group 15 data-repair release `20261104-v2`, 2,452 days, with repair log; accessed 4 November 2026.
[2] Brief and teacher clarifications, 2026: asset specifications, 10 t/day offtake cap, discount convention.
[3] Office for National Statistics, Consumer Prices Index including owner occupiers' housing costs, series D7BT, monthly; all money rebased to 2020 GBP.
[4] BEIS, *Hydrogen Production Costs 2021* (real 2020 GBP); IEA, *Global Hydrogen Review 2025*; ITM POSEIDON data sheet: electrolyser capital, fixed O&M, stack renewal, 55.9 kWh/kg.
[5] Danish Energy Agency, *Technology Data for Energy Storage* (2020 prices, 2025 technology): installed cost, component lifetime, O&M. Its 0.288m EUR/MWh is the value the brief adopts, so the brief and the source coincide here.
[6] DESNZ, *HAR1 successful projects*, December 2023: weighted strike price £241/MWh (£175/MWh in 2012 prices). Hydrogen netback after non-power variable costs, rebased to 2020 money.
[7] Codex Tier 1 v2, an independently written Xpress build in the Tier 1 workspace, commit `71ce76e`; the cross-implementation result is reported in Section 7.